

THE OHIO Naturalist

PUBLISHED BY
THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY

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Volume 1.

February, 1901

Number 4

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PRESS OF HANN & ADAIR

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No. 4, February 1901**

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*** START OF THE PROJECT GUTENBERG EBOOK THE OHIO
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The Ohio Naturalist

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FEBRUARY, 1901

No. 4

THE CORNING OIL AND GAS FIELD.^[A]

J. A. BOWNOCKER

AREA.—This field lies in the three counties, Athens, Perry and Morgan. Leaving out of consideration at present a few small outlying pools, and starting at the south, the productive territory may be said to begin in Section 22, Trimble township, Athens county. From this point it runs almost due north to the Perry county line. The widest part of this portion of the field does not exceed one-half mile, while the narrowest portion permits of a single row only of wells.

^[A] Published by permission of Edward Orton, Jr., State Geologist.

It enters Monroe township, Perry county in Section 33, and runs almost due north towards Corning, but bends to the northeast about one mile south of that place. Here the productive territory attains a maximum width of 3 miles, the greatest in the field. The northeast course is continued to the Morgan county line, where it turns due north, skirting that line with a productive strip about one-half mile wide for 2½ miles, when it turns slightly to the east, entering Morgan county in Section 31 of Deerfield township. From that place it extends through Section 30 and into Section 19, beyond which it has not been traced. Development of this part of the field is retarded by floods of salt water which may limit it in this direction. The total length of the field is about 14 miles.

Outside of this belt are four pools, two of which are of little importance. One lies around Glouster, and has an area of less than one square mile. A second one lies in Sections 22 and 29, a short distance northwest of Glouster. This is the principal territory from which the gas of Corning and surrounding towns is derived. A third pool lies around Porterville, and the fourth known as the Oakfield lies from 3 to 5 miles north of Corning. It includes parts of sections 5, 21, 28, 29, 32, and 33 of Pleasant and 22, 27

and 34 of Bearfield townships. It is in this pool that the most extensive work is being done at the present time.

DISCOVERY.—Probably the first deep well drilled in the Sunday Creek Valley was near Burr Oak, about 4 miles south of Corning. Its date is not now known, but it must have been 40 or more years ago. Its depth is likewise unknown, but it is reported to have penetrated the salt sand. To this day it flows salt water, and with it sufficient gas to be ignited. This well, however, seems not to have aroused suspicion that there might be valuable liquids other than salt water buried in the rocks.

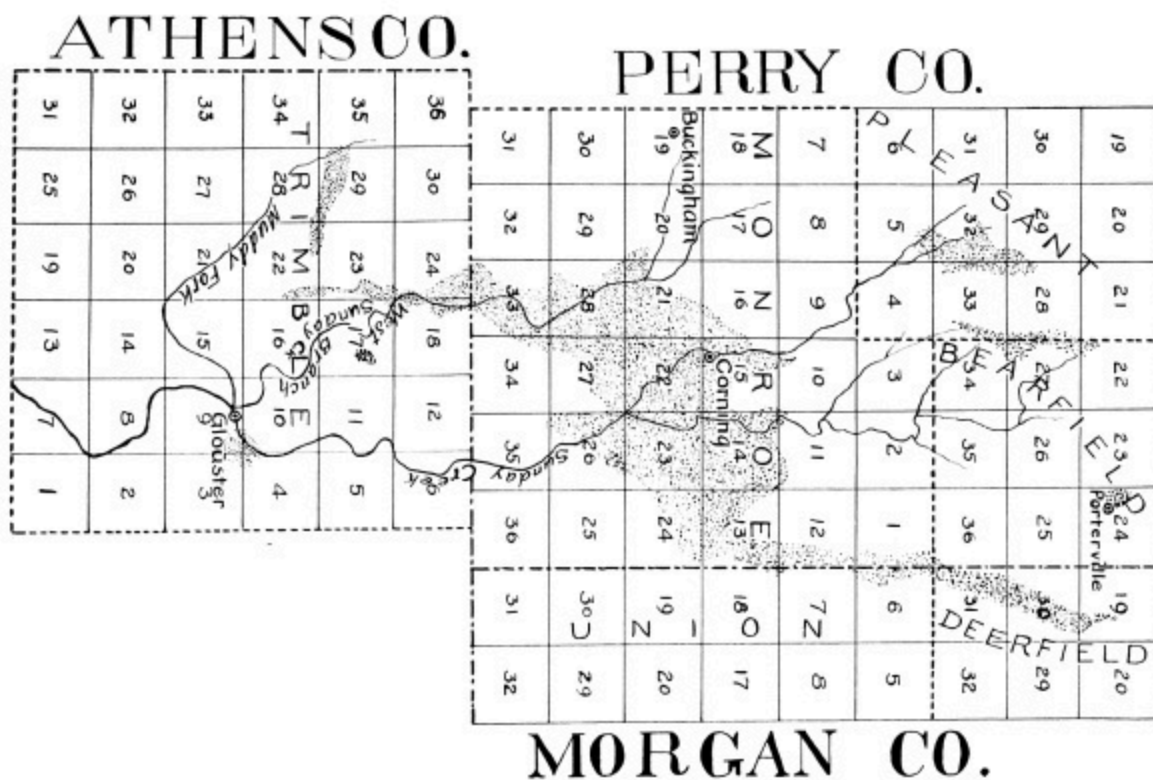
The discovery of oil in the Corning field was a matter of accident, and resulted directly from a scarcity of water for the Toledo and Ohio Central railroad. To remedy this a deep well was drilled in August, 1891, at the round-house, about three-fourths of a mile south of Corning. The only water found was in the salt sand which is reported as having been struck at a depth of 630 feet. The supply was copious, but the salinity prevented its being used in locomotives. This brine was shut out of the well by casing and the drill forced down to a depth of 1507 feet. Finding no water at that depth the work ceased, but a few days later oil was thrown to the top of the derrick, and there were smaller eruptions later. However further disturbances of this sort were prevented by the company closing the well.

DEVELOPMENT.—The disclosure made by this well attracted the attention of oil men who immediately entered the field and began leasing territory. The citizens of Corning feared the territory was falling into the hands of the Standard Oil Company, and that it might not under such conditions be developed for years. Accordingly a home company styled "The Sunday Creek Oil & Gas Company," was organized in February, 1892, to make certain the development of the territory. The capital stock was placed at \$10,000 in shares of \$50, and \$8900 of the stock was sold. Much of this was raised by citizens of the town subscribing for single shares.

The new Company was successful. By January 1st, 1898, 255% in dividends had actually been paid the stock holders. In September, 1898, a power for pumping the wells, and costing over \$7000, was erected, the contractor taking the product of the wells until it paid for the plant. In November, 1899, the property together with \$1250—the amount received in excess of the cost of the plant—was turned over to the original holders. The

power is now (July 1, 1900,) pumping 20 wells, which have a daily production of 40 barrels.

The first well drilled by this Company was on the William Fisher farm in northwest quarter section 14, Monroe township, Perry county. The Berea was struck at 1012 feet, but the indications were so unfavorable for a paying well that it was not considered advisable to shoot it. However, on June 2d, 1892, after waiting nearly a month, the well was shot with 80 quarts of nitro-glycerine, which had been hauled from Sistersville, W. Va. The cost of the shot was \$200. The first day following the shooting of the well it produced 12 barrels, and a year later was still producing 10 barrels per day. Following this other wells were drilled in sections 14 and 15. In all 25 have been drilled, only 3 of which were dry holes.



THE CORNING OIL *and* GAS FIELD

BY J.A. BOWNOCKER.

[Click on image for expanded map.]

Other companies began work and the territory was rapidly leased and tested. Naturally operations began near the round-house where oil had first been shown to exist. From this as a center the drill moved out in all directions until the limits of the field had been disclosed. The later work has been along the northeast end of the territory in Morgan county, where the oil seems to be shut out by reservoirs of salt water. During the present summer (1900) the valuable pool in the Oakfield district has been developed, though small wells had been found there several years earlier. The principal farms are the Porter, Longstreth, Donnelly, Monahan, McDonald and Grenen. The first well was on the Porter farm and was finished early in 1900. Its production was 35 barrels the first day. The second well was on the Monahan farm. It was completed soon after the Porter well and had an initial flow of 45 barrels in 24 hours. The next two wells were drilled on the Longstreth farm, and both were fair producers. Early in the Spring a well was completed on the Donnelly farm and flowed 125 barrels the first day. Other wells on this farm are much smaller. Two wells on the Grenen farm began flowing 675 and 90 barrels respectively. It is interesting to note that the development of this, the richest part of the Corning field, occurred late in the territory's history. Possibly other pools of equal richness may yet be discovered lying near the principal field.

An important step in the development of the field occurred on August 13th, 1893 when the Buckeye Pipe Line was completed. Before that the oil was transported by tank cars. The oil which is brought to the tanks partly by gravity and partly by suction, the latter being produced by an 8 horse-power gas engine, is stored in two iron tanks, one of which has a capacity of 30,000, and the other 28,000 barrels. From these tanks the oil is forced to Elba, a distance of 34 miles, through a 4 inch line. This work is done by a 35 horse-power engine which gives a pressure in the line of from 700 to 1000 pounds per square inch. The rate at which the oil is transported varies with the temperature. In the summer when the oil is warm, and hence thin, 128 barrels may be pumped in one hour, but in the winter when the oil is cold and thick the transportation may be restricted to 11 barrels for the same period.

When the pipe line was completed the production of the field was about 500 barrels per day. It increased to 1300 barrels in 1896, but since then has

declined. At present it ranges from 800 to 900 barrels per day. The total production of the field is shown by the following letter:

The Buckeye Pipe Line Company—Macksburg Division.
Oil City, Pennsylvania, October 25th, 1900.

J. A. Bownocker, Esq., Columbus, Ohio:

Dear Sir—Your favor of October 18th to Superintendent N. Moore, asking for the total production by years of the Corning Field, has been referred to me. Below please find the figures of oil received by The Buckeye Pipe Line Company from the Corning Field from August, 1893, to September, 1900:

Part of Year 1893	128,918.03 Bbls.	
Year 1894	322,313.71	“
Year 1895	428,385.03	“
Year 1896	469,258.78	“
Year 1897	328,188.11	“
Year 1898	196,417.75	“
Year 1899	211,060.22	“
January 1st to July 31st, 1900	143,314.96	“
August, 1900	26,929.66	“
September, 1900	22,517.67	“
Total	<u>2,277,303.90</u>	Bbls.

July 31st, 1900, completes the first seven years production and thinking you might prefer to use the even years, I have given you the figures for the year 1900 to July 31st in one lump and the oil taken from that field for the months of August and September separately.

Trusting this will answer your purposes, I remain

Yours truly,
J. R. CAMPBELL, Treasurer.

LEASES.—At first the operators paid no bonuses, but gave a royalty of one-eighth of the oil to the land owners—a rate of compensation that has been usually maintained. To this there is one exception worthy of note. When the round-house well showed the existence of oil, and operators began leasing the surrounding territory, Fredrick Weaver, a thrifty German farmer residing a short distance east from the round-house, quietly visited the oil fields of Washington, Pennsylvania, and investigated the methods of leasing oil territory in that field. When he returned home he demanded a royalty of one-fourth the oil and a bonus of \$200 for each of the eight wells which it was proposed should be drilled on his farm of eighty acres, and since his territory was regarded as very promising, these rather severe terms were granted. However, after drilling six wells, and the territory not meeting expectations, the contractors complained and Mr. Weaver generously reduced the bonus. More recently a royalty of one-sixth the oil has been received by holders of lands that were deemed especially promising, and bonuses also have been received. The leases usually required that a well be drilled in from thirty to sixty days, but sometimes, especially in the least promising territory, six months were allowed.

That the field was a monopoly for no one is shown by the following list, which includes the chief operators of the district:

- Corning Oil Company.
- Denman & Thompson.
- O'Connel Oil Company.
- Brooks Oil Company.
- Caldron & Snyder.
- Sunday Creek Oil & Gas Company.
- Perry County Oil Company.
- W. B. Barker & Company.
- Cleveland Oil Company.
- Keystone Oil Company.
- W. E. Detlor.
- William Rosier.
- J. H. Van Wormer.
- Northeast Oil Company.
- Becker Oil Company.
- Allen, Sternberg & Company.

Bolivar Oil Company.
William McMullen.
A. Bulger & Company.
Corning Natural Gas Company.
L. D. Langmade.
Harrington Brothers.
Ohio Oil Company.
Church Oil Company.
Monroe Oil & Gas Company.
Weaver Brothers.
W. B. Irwin & Company.
George Best & Company.
Foster & Moran.
Fallen Rock Company.
Ohlviler & Chambers.
Mill Oil Company.
Hemlock Oil Company.
John Holden.
Wells & Foraker.
Longfellow & Stevens.
Russell Metzger.
McGee & Stewart.
Stratton & Mark.
National Oil Company.

GEOLOGY OF THE REGION.—The surface of the territory lies in the Lower Productive and Lower Barren coal measures. The highest hills reach up to or extend above the Ames or Crinoidal limestone. In fact along the northeast extremity of the field the hills are capped by the limestones which underlie the Pittsburgh coal. The deepest valley—that of Sunday Creek—cuts through the Middle Kittanning coal, a short distance north of Corning, but at this town the seam named is under cover, while the Upper Freeport coal is at about drainage level.

The succession of strata under ground is shown by the following record kept and furnished the Survey by Mr. G. W. Delong, Superintendent of Schools, Corning. The well is located on lot 154 of the town just named, and the top of the well lies at the base of the Mahoning sandstone:

	Thickness of Stratum		Total Thickness	
Shale	25	feet	25	feet
Bastard Lime	15	“	40	“
Sand	10	“	50	“
Coal (No. 6)	10	“	60	“
White Slate	65	“	125	“
Sand	15	“	140	“
White Slate	25	“	165	“
Blue	10	“	175	“
Sand	10	“	185	“
Slate	50	“	235	“
Shale	35	“	270	“
Sand	30	“	300	“
Black Shale	10	“	310	“
Lime	25	“	335	“
Shale with Concretions	100	“	435	“
Slate	25	“	460	“
Limestone(?)	30	“	490	“
Shale	35	“	525	“
Salt Sand	30	“	555	“
White Slate	100	“	655	“
Slate and Concretions	25	“	680	“
Shale	15	“	695	“
Little Salt Sand	20	“	715	“
White Slate	100	“	815	“
Slate and Concretions	100	“	915	“
Brown Shale	40	“	955	“
Black Shale	38	“	993	“

	Thickness of Stratum	Total Thickness
Top Berea		993 “
Bottom of Berea		1008 “

The depth of the well as shown by the steel line is 1012½ feet. It was drilled in the Fall of 1896, and was shot with twenty quarts of nitro-glycerine. It began flowing thirty barrels per day, but the production has diminished until at present it is producing only one barrel per day. Below the Berea the Bedford shales are found in their normal conditions.

THE OIL SAND.—This is in all cases the Berea. The sand has the light gray color so common in this formation in other parts of the state. It is moderately fine grained, but there is considerable variation in this respect. Usually it is a pure quartz sand, but occasionally has thin layers of dark shaly material running through it. In thickness it shows considerable variation, but never disappears in this field. The normal thickness is usually given as twenty feet and the maximum reported is eighty feet. This depth was found on the Potts farm about one and one-fourth miles northeast of Corning, and on the O’Farrell farm about two miles east from the same town. In both cases a dark gray shale, probably the Ohio, lay below. The Bedford on this theory had been swept away before the Berea was deposited. In such abnormal depths the additions always appear to be on the bottom, showing that the surface of the underlying Bedford shale was quite uneven. Here, as elsewhere in the state, the drill shows the upper surface of the Berea to be uniform. It is worthy of note that the production of oil does not vary as the thickness of the sand. In fact in this field the great thicknesses are generally poor producers.

The “pay streak” or that containing the oil and gas ranges in thickness from 3 to 8 feet, but very few of the wells attain the maximum figure. Towards the margin of the productive field the “pay streak” thins, and finally disappears. The top of the “pay” usually lies from 10 to 15 feet below the surface of the Berea. As a rule the “pay” is coarser than other parts of the Berea, and generally the coarser the rock the larger the well. Sometimes in the thick part of the Berea there are two “pay streaks.”

THE WELLS.—The number of wells producing July 1, 1900, exceeded 600. About 100 dry holes have been drilled and about an equal number of wells have been abandoned, so that 800 is a fair approximation of the total number of wells drilled in the field. As a rule a well has been put down for each 8 to 10 acres of surface territory.

The wells have been cased through the salt sand, a depth of 555 feet in the valley at Corning. The casing has almost invariably been 5 $\frac{5}{8}$ inches, inside measurement. The rocks comprising the underlying 160-180 feet, and terminating with the "Little Salt Sand" have furnished some water which has been disastrous to the wells. It reduced the gas pressure, thus necessitating pumping the wells earlier than otherwise would have been required, and perhaps prematurely destroying the life of the well. Had the wells been cased through the "Little Salt Sand" time and money would have been saved, and the production of the field would have been larger.

The western side of the field is quite free from salt water. It is on that side that the principal gas territory lies. On the eastern side of the field the conditions are more variable. In Trimble township, Athens county, the wells are free from water, while in Monroe township, Perry county, salt water is found in the northeast corner, and in Morgan county it is so abundant that operating is prevented. From this it appears that the western side of the Corning field is free from salt water, and that it is absent also on the eastern side at the southern margin of the territory, but that it increases rapidly to the northeast.

While the production of the wells after being shot has varied greatly, yet they have not furnished the great extremes that many other fields have. Few, if any of the wells, have started better than 125 barrels per day, and it has been estimated that the average for the entire field has been 20 barrels.

The wells have sufficient gas pressure to flow them during the earlier part of their lives, but later as the pressure diminishes they have to be pumped. Since the eastern side of the field has salt water the wells there have to be pumped earlier than those on the western side.

THE GAS WELLS.—The principal gas territory is that along Muddy Fork in Sections 22 and 29 Trimble township, Athens county, the best wells being found in the western half of the latter section on the lands of the Hocking Coal and Railroad company. The largest well in this field started at

3,000,000 cubic feet per day with a rock pressure of 400 pounds. It was drilled in the fall of 1897, and one year later was producing 2,000,000 cubic feet per day, and still another year later 1,500,000 cubic feet. Of the other wells in this territory two started at 2,000,000 feet each, two at 1,000,000 feet each, and three at 500,000 feet each. The decline in the smaller wells was not as rapid as in the larger ones since the demands made on them were not as heavy. Thus far no dry holes have been found in this territory. The reliance of the community is on this field where 5,000 acres are leased in one block.

Another district that has yielded considerable gas is that at Oakfield about 3 miles north of Corning. These wells started at 2,000,000, 1,500,000, 500,000, and 250,000 cubic feet per day respectively. Two of the smaller of these have been abandoned after having produced for two years. The largest of these wells, now four years old, is producing 500,000 cubic feet per day, and the second largest, now three years old, is producing the same amount. The wells in this field produce considerable oil and by some are rated as oil wells rather than gas ones.

Outside of these two places an occasional strip is found that produces gas in paying quantities. Thus about one mile northeast of Corning two wells were drilled, which combined produced 500,000 cubic feet per day. They produced three years and were then abandoned. About two miles north of Corning a good well was drilled on the Newberry farm. It started at 1,500,000 cubic feet per day, had an initial rock pressure of 400 pounds and lasted three years.

Another productive tract lies about 6 miles northeast of Corning on the Finley, Devore and Stoneburner farms. Three wells were drilled on the Finley farm, and started one at 1,500,000 and two at 250,000 cubic feet, with an initial rock pressure of 400 pounds. These wells lasted three years.

The operators of the wells have been much troubled with salt water in the Muddy Fork field and with oil in the Oakfield territory. Salt water is removed by "blowing" the wells. For this operation the wells are closed for a short period, usually about 30 minutes, allowing the gas pressure to increase; when this has become sufficiently strong the well is opened at the top and the gas then blows the water from the well. When the well has been cleansed in this manner it is closed and the gas turned back into the mains.

Sometimes, however, the weight of the water is so great that the gas cannot drive it from the well in the manner just stated, especially is this true with wells that have been in use for a considerable period. Then an iron rod attached to a long pole is let down through the water, is raised and lowered, and the gas following the pole in its ascent finally drives the water from the well. This method of cleaning is known as “agitating.” Finally the pressure of the gas becomes so small that it cannot lift the water with the help of “agitating,” and then the well is dead. In winter time each well is cleaned every other day, and in the summer twice a week.

The gas wells in the Corning field are owned and operated by the Corning Natural Gas Company. It supplies Jacksonville, Trimble, Glouster, Murray City, New Straitsville, Shawnee, Hemlock, Corning, Rendville, Moxahala, New Lexington, and several interior hamlets. Almost the sole use of the fuel is for heat and light.

The company makes a rate of 20 cents per thousand feet by meter. Where the meter is not used, the prices in winter are \$2.00 per month for the first fire; \$1.50 for the second; \$1.00 for the third; 75 cents for the fourth, and all additional fires at the latter figure. In the summer a charge of \$1.50 for each cooking fire is made. For lights the charges are 25 cents each for the first two and 15 cents for each additional one.

The number of customers supplied by this company in 1900 was approximately as follows:

Corning	300
Glouster	250
New Lexington	250
Shawnee	200
New Straitsville	150
Other places	350
Total	<u>1500</u>

In the Fall of 1899 the wells of the company produced 6,000,000 cubic feet per day, but during cold weather when the demand for fuel was great they dropped to 3,000,000 cubic feet, and the rock pressure which was 300 pounds in the Fall was only 200 during the winter. On July 7th, 1900, the

rock pressure of the wells in the Muddy Fork field ranged from 170 to 280 pounds, indicating a considerable drop from that of the preceding autumn. The company expects to drill four additional wells during the ensuing fall (1900) in the Muddy Fork territory, and by so doing expects to keep three wells closed, and thus maintain a good rock pressure.

TWELVE ADDITIONS TO THE OHIO PLANT LIST.

W. A. KELLERMAN.

The species named below have not been reported in the Fourth State Catalogue of Ohio Plants, in the First Annual Supplement, nor in "Additions to the Ohio Flora," O. S. U. NATURALIST, 1:15. The serial number prefixed to each name indicates where in the Fourth State Catalogue the species should be inserted. The first collector and locality are given for each of the listed specimens.

- 82a. *Potamogeton nuttallii* Cham. & Schw., Nuttall's Pondweed. Donor's Lake, Wayne County, Ohio (Reported by A. D. Selby before meeting of Ohio Academy of Science, December 27, 1900.)
- 488a. *Juncus dichotoma* Ell. Forked Rush. Cuyahoga County. (Edo Claassen.)
- 647a. *Betula alba pendula* Hortorum. Pendulous White Birch. Escaped; Painesville. (Otto Hacker.)
- 781a. *Dianthus deltoides* L. Maiden Pink. Escaped: Painesville. (Otto Hacker.)
- 936a. *Erysimum repandum* L. Logan County. (A. D. Selby in report before Ohio Academy of Science, December 27, 1900.)
- 1117a. *Lespedeza angustifolia* (Ph) L. (*L. capitata* var. *angustifolia* Ph.) Narrow-leaf Bush-clover. Fulton County. (A. D. Selby in report at meeting of Ohio Academy of Science, December 27, 1900.)
- 1211a. *Rhamnus caroliniana* Walt. Carolina Buckthorn. Adams and Brown Counties. (W. A. Kellerman.)

- 1526a. *Teucrium scorodonia* L. European Germander. Escaped; Painesville, Lake County. (Otto Hacker.)
- 1650a. *Gerardia besseyana* Britt. (*G. tenuifolia* var. *macrophylla* Benth.) Bessey's *Gerardia*. Wooster. (J. W. T. Duvel; reported by A. D. Selby at meeting of the Ohio Academy of Science, December 27, 1900.)
- 1759a. *Leontodon hastilis* L. Dandelion. Escaped; Painesville, Lake County. (Otto Hacker.)
- 1766b. *Lactuca saligna* L. European Wild Lettuce. Dayton. (A. D. Selby.)
- 1777a. *Hieracium greenii* Porter & Britt. Green's Hawkweed. Wayne County. (Selby and Duvel; reported by A. D. Selby before meeting of Academy of Science, December 27, 1900.)

There were 2025 species reported in the Fourth State Catalogue (1899) for the State of Ohio. This number was supposed to be approximately correct, since those of previous lists were discarded which are known to have been erroneously identified or were unquestionably beyond our range. In the First Annual Supplement sixty-nine additions were made, and in Additions to the Ohio Flora, (*O. S. U. NATURALIST*, 1:15) twenty-two more were recorded. Therefore those enumerated above bring our grand total to 2128 species of Pteridophytes and Spermatophytes.

A NEW SPECIES OF GOMPHUS AND ITS NEAR RELATIVES.

JAMES S. HINE.

Gomphus viridifrons n. sp. Length of the abdomen about 33 mm., hind wing about 27 mm.; black, face and occiput green; prothorax with anterior margin and three spots green or yellow; thorax green with spaces at base of wings, lateral suture and six bands before black, the two middle bands are abbreviated anteriorly and separated by the mid-dorsal carina which is very feebly green. Abdomen black, a dorsal band and sides of first two or three segments yellowish, a yellow spot at base of each of segments four to seven, and sides of eight and nine usually yellowish.

Abdominal appendages of the male straight, about as long as the tenth segment, from above, widest at base, gradually narrowed from apical third and acute at apex; from the side prominently widened at base, with a strong tooth beneath at two-thirds of the length. Hamules large, of nearly the same width for the whole length and ending behind in a hooked process. Vulvar scale almost as long as the ninth abdominal segment, gradually narrowed, apical third divided and the two parts divaricate.

Described from fourteen males and a female taken at Loudonville, Ohio, June 14, 1900; and a male and female taken at Ohio Pile, Pa.—the latter two specimens by E. B. Williamson.

The species averages larger than either *brevis* or *abbreviatus* and may be separated from the former readily by its green face, by the striking differences in the vulvar lamina and by the hamules and male appendages. In *brevis* the tooth on the appendage is nearer the end and the space from it to apex is noticeably curved while in *viridifrons* this space is practically straight.

It has more points in common with *abbreviatus*, but in that species the vulvar scale is short and triangular, the hamules are smaller and shorter, and

the tooth on the superior appendage of the male is much farther from the apex.

Through the kindness of Dr. Calvert and Mr. Williamson I have at my disposal, specimens from which many of the accompanying drawings were made. In *viridifrons*, *brevis* and *abbreviatus* I have made drawings from different specimens of the same species to show slight variations.

This is *Gomphus* sp. Williamson, Dragonflies of Indiana, 294.

A COMPARATIVE STUDY OF GOMPHUS FURCIFER AND VILLOSIPES.

JAMES S. HINE.

It seems that nearly every author who has considered these two species has compared them. Considering general appearances they are much alike but can be separated easily by several details and as I have good material of both sexes of the two species before me, I thought it might be of consequence to consider in a comparative way some of their characteristics.

The occiput in both sexes of *villosipes* bears a prominent tooth at the middle of its upper edge. This tooth varies some in different specimens, it ends above in a single point or it may be widened and end above in three or more points. In *furcifer* there is no tooth on the occiput in either sex.

The vulvar scales are very different, in *villosipes* the part may be said to be triangular with the free sides curved and the apical part divided for about half the length of the scale. In *furcifer* its basal part is similar but the apical part is produced, giving quite a different form. In the former species the scale is about one-third as long as the ninth segment, while in the latter it is about three-eighths as long as that segment.

The abdominal appendages of the male are different. From dorsal view those of *furcifer* are rectangular with the inner distal angle very much produced inward and backward; in *villosipes* they are wide at base, oblong, with the outer distal part broadly rounded and the corresponding inner part produced directly backward. The hamules are characteristic and may be explained best by reference to the figures.

The two species are colored much alike but *furcifer* is darker. The tenth abdominal segment may be said to be yellow in both but in *furcifer* the sides of the segment are dark, oftentimes black.

They agree in habits, both preferring to fly over stagnant water where the males come to rest on floating objects or on the ground at the water's edge. I have observed the female of *furcifer* ovipositing in stagnant water among lily pads and other aquatic plants. Her flight is slow while thus engaged, and her actions more like a *Libellula* than the usual *Gomphus*.

EXPLANATION OF [PLATE](#).

Gomphus villosipes—1, dorsal view of male appendages; 3, vulvar scale; 5, hamule; 7, occiput of female; 8, occiput of male.

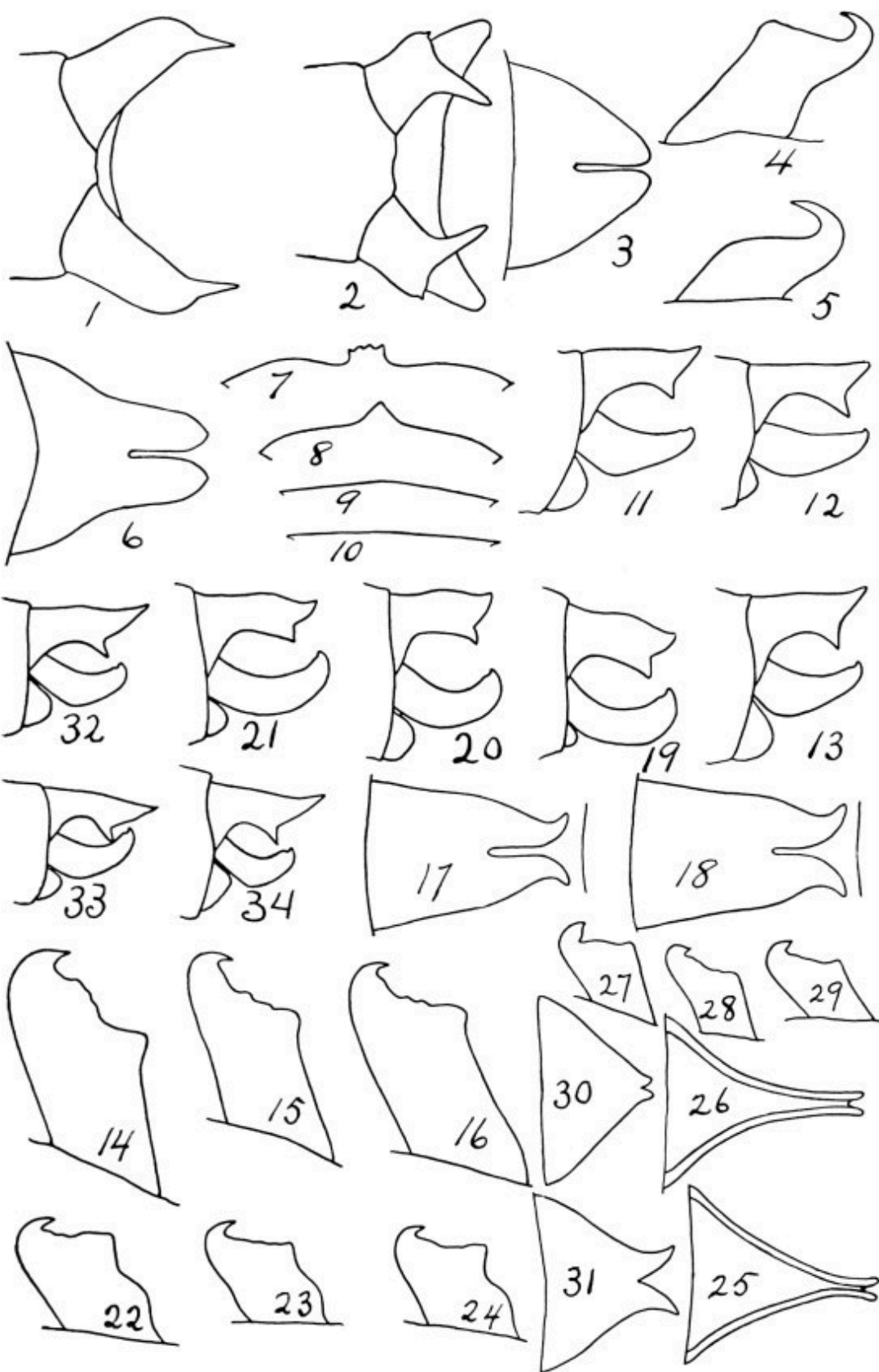
Gomphus furcifer—2, dorsal view of male appendages; 4, hamule; 6, vulvar scale; 9, occiput of male; 10, occiput of female.

Gomphus viridifrons—11, 12, 13, dorsal view of male appendages—13 from Pa., 11, 12 from Ohio; 14, 15, 16, hamules; 17, 18, vulvar scales—18 from Pennsylvania, 17 from Ohio.

Gomphus brevis—19, 20, 21, dorsal view of male appendages—specimens from Maine and Pennsylvania; 22, 23, 24, hamules; 25, 26, vulvar scales.

Gomphus abbreviatus—27, 28, 29, hamules; 30, 31, vulvar scales; 32, 33, 34, dorsal view of male appendages—specimens all from Maine.

Plate 5.



HINE—GOMPHINE STUDIES.

NOTES FROM BOTANICAL LITERATURE.

W. A. KELLERMAN.

A very interesting Bulletin of eight pages on the Chrysanthemum Rust has been issued by J. C. Arthur, botanist of the Indiana Agricultural Experiment Station. It is No. 85, and is dated October, 1900. Dr. Arthur gives a general account of Uredineae or Rusts, explaining that the typical forms have three prominent sets of spores, namely (1) aecidiospores, usually red or orange in little white cups, (2) uredospores, generally of a rusty yellow color and abundant (hence the group name, Rusts), and (3) teleutospores, or the Winter spores, usually dark brown or black. He further states that only uredospores have yet been found in Europe and America, and since the fungus is an annual, it is puzzling to see how it escapes extermination in winter and spring when Chrysanthemum plants are latent.

The assumption that this was the well known and common *Puccinia hieracii* or *Puccinia tanacetii* he proves to be incorrect by inoculation experiments. Uredospores from Chrysanthemums he sowed on Chrysanthemums and obtained a crop of uredospores. Similar uredospores sowed on Dandelion, Burdock, and Ox-eye Daisy produced no infection; uredospores from the latter hosts sown on Chrysanthemum likewise produced no infection. Uredospores from Dandelion sown on Dandelion produced uredospores. Others have tried similar experiments, using Tansy, Costmary, Orange Hawkweed, Giant Daisy, and Marguerite, besides the host plants named above, but the Chrysanthemum Rust refuses to grow on any of them. This Rust, which is common and well known on the Chrysanthemum in Japan, has been named *Puccinia chrysanthemi* by Rose.

In connection with suggestions relative to combating the disease, Dr. Arthur says that "so long as the teleutospores do not make an appearance in this country, the careful cultivator may feel assured that a moderate amount of timely effort will enable him to rid his establishment of the Rust, if he is

so unfortunate as to have it donated to him by some careless florist. Observations made by the writer and others show that the tendency is for the disease to disappear of itself, to run its course in an establishment and die out, which is very likely to some extent due to the absence of teleutospores.”

BRYOLOGY.—Mrs. Britton’s popular articles on the Mosses and how to study them, that have appeared from time to time, furnished the directions and incentive to many who before had taken little or no interest in this group of plants. Her purpose and plan are imitated and extended in a charming little book, that has been prepared and published by Dr. A. J. Grout, of the Boys’ High School, Brooklyn, New York, called “Mosses with a Hand-Lens.” The author says that many years of study of Mosses in the field and in Herbaria have convinced him that “any person of average intelligence can easily learn to recognize seventy to one hundred common mosses, with the aid of a hand-lens of ten to fifteen diameters magnifying power.”



Fig. 9, Ceratodon.

I have Dr. Grout's permission to reproduce two illustrations, which represent fairly that phase of his valuable book. [Figure 9](#) shows *Ceratodon purpureus* L, and the text pertaining to the same is as follows: "Ceratodon is one of the commonest of all our mosses. It is found on the edges of paths, roofs of old buildings, sand by the seashore, and in general any barren compact soil is its favorite habitat. The plants are short and grow close together, forming dense thin mats of dark green. The lance-like young sporophytes appear early in spring as soon as the snow is melted. By the middle of summer the capsules often decay beyond recognition, and the seta breaks from the plant at the touch.

Unless one has become very familiar with *Ceratodon* it is not always easy to recognize it without mature capsules. When the capsules have fully matured they shrink when dry and become furrowed. This peculiar furrowing, the dark rich color of the capsules, a color called purple by the older botanists, but which is really a very dark chestnut or red-brown, make it easy to recognize this species.”

[Plate II.](#) shows the Hair-cap Mosses, *Polytrichum*, the largest of all our species. There are four common species all having square capsules which character distinguishes them from *Pogonatum*s, the latter having cylindric capsules. “The Ohio Hair-cap without the sporophyte (seta and capsule, commonly called fruit) is not readily distinguished from the Common, as the leaves and general appearance are very similar. But with the sporophyte present, the distinctions are clear. In Figs. b and d ([Plate II.](#)) note that the capsule of the Common Hair-cap is almost cubical, that the lid has a *very short beak*, and that the capsule is entirely covered by the calyptra. The capsule of the Ohio Hair-cap (e) is elongated, slender with a tapering neck, and with a much longer beak to the lid. The lid and the calyptra of the Ohio Hair-cap fall early in June, very soon after the spores are ripe, and it is not always easy to find either in position, but if the calyptra be found, it will be seen to cover the upper portion of the capsule only. The Common Hair-cap, although occurring in woods, is most common in open fields; the Ohio Hair-cap being most frequent in shady, more moist spots, often in deep woods. The remaining two species are easily distinguished from the two mentioned above by the margins of the leaves, which are thin and membranaceous, and are folded in over the central portion of the leaf, [as illustrated](#) in o, o' and p.”

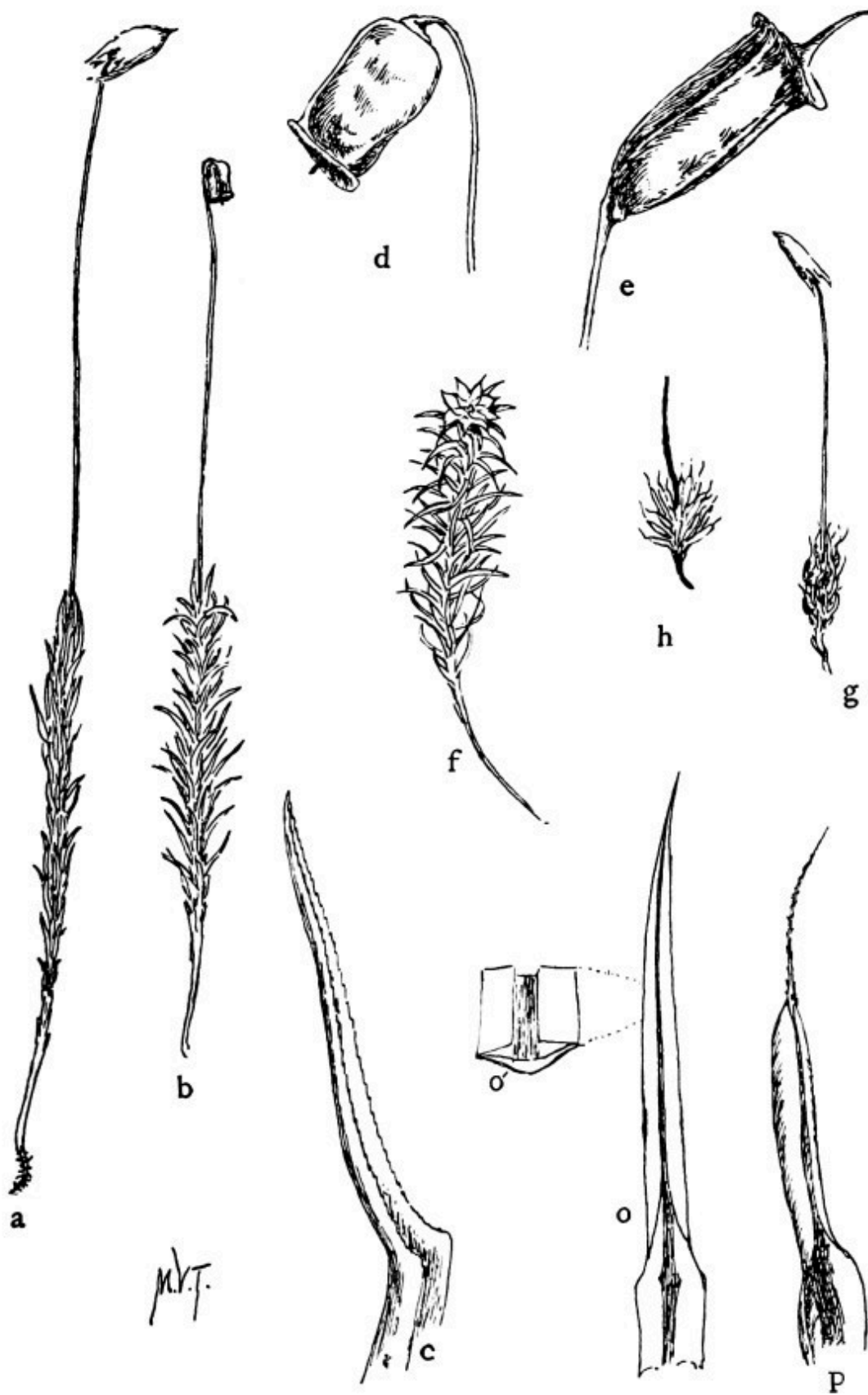


Plate II., Polytrichum or Hair-cap Moss.

Figs. a (dry), b (moist), c (leaf),
d (capsule) and f are *P. commune*;
Fig. e, capsule of *P. ohioense*;
Figs. g, h and p, *P. piliferum*;
Figs. o and o', *P. juniperinum*.

I can not too strongly commend "Mosses with a Hand-Lens" (price \$1.10) prepared with the purpose of giving "by drawings and descriptions the information necessary to enable any one interested to become acquainted with the more common mosses with the least possible outlay of time, patience and money." The book contains a key to the genera based mainly on structural characters and one mainly on habitat, also many keys under the genera; copious illustrations, clear and accurate on almost every page of the text and eight full-page plates from drawings by Mary V. Thayer; and an illustrated glossary of bryological terms.

MEETING OF THE BIOLOGICAL CLUB.

The Biological Club met in Zoological lecture room December 3, 1900. Prof. Herbert Osborn presided, twenty-six members present. The following papers were presented:

“Notes on the Saw Brier, and a *Rhamnus* new to Ohio.”

“The Waverly Series of Ohio.”

In the first paper Prof. Kellerman spoke of the distribution of the Saw Brier, *Smilax glauca*, in the southern part of the State, and exhibited specimens showing its striking variations in form of leaves.

The southern buckthorne, *Rhamnus caroliniana*, was observed commonly in Adams County. It also occurs in Brown County. This is the first record for this species in Ohio. Specimens in fruit were exhibited.

A hackberry was found unlike any form hitherto reported from Ohio. Specimens are in the hands of Rev. E. J. Hill for study and determination.

Prof. Ball reported leaf variation as occurring commonly in Colorado, and that various leaf forms could be observed in climbing a single hill.

Prof. Prosser, in the second paper, reviewed the literature that has been published on the Waverly Series of Ohio, and as a summing up gave a list of names with authorities to be used in future in speaking of the formations of this series. These are as follows:

Beginning above the Huron Shale. 1. Bedford Shale (Newb.) 85 feet. 2. Berea Grit (Newb.) 40 feet. 3. Sunbury Shale (Hicks) 10-15 feet. 4. Cuyahoga Shale (Newb.) 275-300 feet. 5. Black Hand Conglomerate (Hicks) 40-100 feet. 6. Logan Sandstone (Andrews) 115 feet.

Under the head of personal observations, Prof. Schaffner gave a list of trees and shrubs which he and F. J. Tyler had found cutting off (self-pruning) their own branches.

Prof. Kellerman remarked upon the abundance of the red-seeded dandelion in various parts of the state.

Walter Metz, J. A. Beer, H. A. Clark, Charles I. Meade, Miss Elizabeth Sessions, Miss L. D. Wilson, W. P. Simpson, Mrs. J. H. Schaffner, B. B. Wells, Mrs. E. D. Ball, J. N. Frank, A. G. McCall, Miss Carrie R. Weick, A. C. Whitmore, Miss Caroline Meade and Miss Maud Flynn were elected to membership.

Professors Prosser, Landacre and Mr. Griggs were appointed a committee to locate board and lodging for members of the Ohio Academy of Science. Adjourned.

JAS. S. HINE, Secretary.

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The illustrations have been moved so that they do not break up paragraphs and so that they are next to the text they illustrate.

Typographical and punctuation errors have been silently corrected.

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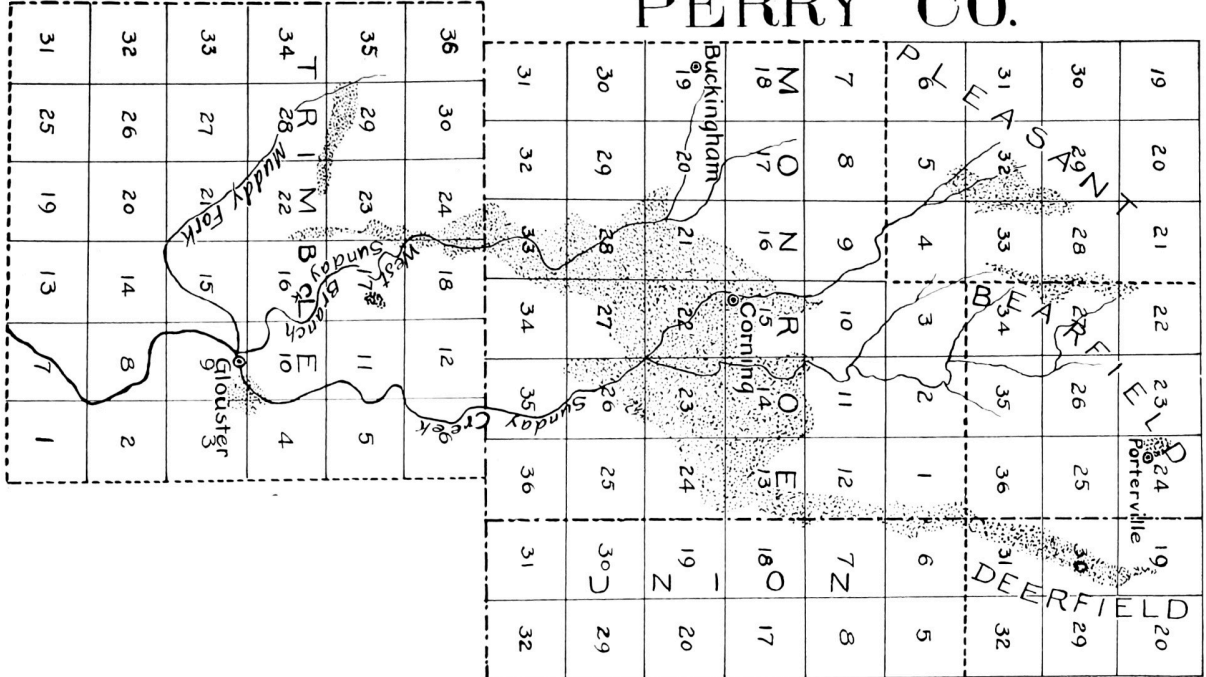
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